# Applied Numerical Methods for Civil Engineering

CGN 3405 - 0002

## Week 8: Introduction to Applied Linear Algebra: Part I

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## Quizzes Now!

- **Today's participation** (ungraded survey): Please check out

  "`Class Participation Quiz 16`"

  Time slot: **2:30PM – 3:00PM**

  on Canvas.

# Linear Systems

- A system of linear equations:

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\
&\vdots \\
a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n
\end{aligned}
$$

# Linear Systems

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- A system of $n$ linear algebraic equations:

$$\sum_{j=1}^{n} a_{ij}x_j = b_i$$

for $i = 1, 2, \ldots, n$.

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- Matrix form $\boldsymbol{Ax} = \boldsymbol{b}$
- **Focus**: The existence and uniqueness of solutions

## Structural Stiffness Analogy

In civil engineering:

$$\boldsymbol{K}\boldsymbol{u} = \boldsymbol{F}$$

- $\boldsymbol{K}$ is the global stiffness matrix
- $\boldsymbol{u}$ is the displacement vector (unknowns)
- $\boldsymbol{F}$ is applied nodal loads

# Matrix Equation

In the field of control theory, how to find unique solution to $\boldsymbol{X}$?

- A Lyapunov equation is a **matrix equation** of the form:

$$\boldsymbol{A}^\top \boldsymbol{X} + \boldsymbol{X}\boldsymbol{A} = \boldsymbol{C}$$

  for any $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{C} \in \mathbb{R}^{n\times n}$

- A Sylvester equation is a **matrix equation** of the form:

$$\boldsymbol{A}\boldsymbol{X} + \boldsymbol{X}\boldsymbol{B} = \boldsymbol{C}$$

  for any $\boldsymbol{A} \in \mathbb{R}^{m\times m}$, $\boldsymbol{B} \in \mathbb{R}^{n\times n}$, and $\boldsymbol{C} \in \mathbb{R}^{m\times n}$

## Linear Transformation

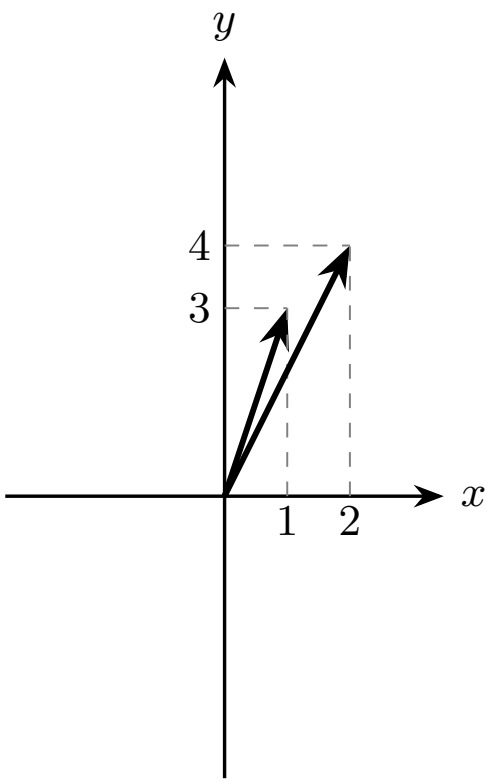
- **Definition**: Just as a linear function $f(x) = ax$ scales a number $x$, a linear transformation $f(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$ maps a vector $\boldsymbol{x} \in \mathbb{R}^n$ to another vector $\boldsymbol{A}\boldsymbol{x} \in \mathbb{R}^n$.
- **Key Difference**: In a linear transformation, the input and output are vectors rather than single scalar numbers.
- **Mechanism**: The transformation occurs by moving the vector $\boldsymbol{x}$ to a new position $\boldsymbol{A}\boldsymbol{x}$ through the operation of matrix $\boldsymbol{A}$.

# Linear Transformation

To understand $\boldsymbol{Ax}$ geometrically, consider $\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\boldsymbol{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

1. **Identify Basis Vectors**: Extract the column vectors of matrix $\boldsymbol{A}$:

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$



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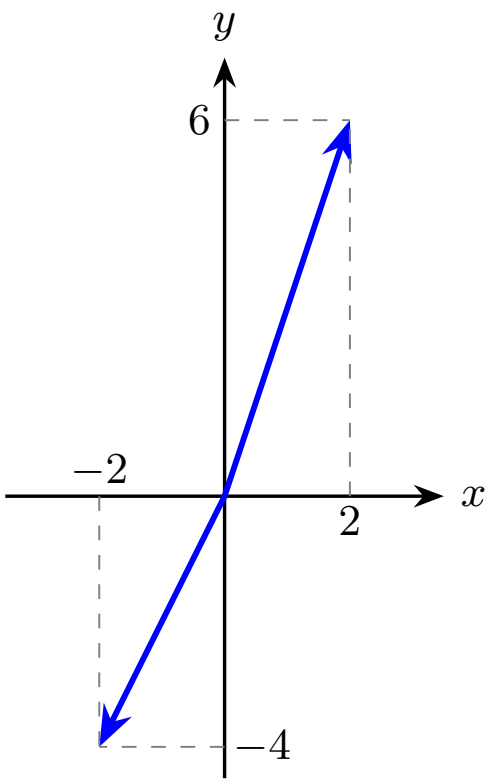
❷ **Scale the Basis**: Multiply each basis vector by $2$ and $-1$:

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} \times 2 = \begin{bmatrix} 2 \\ 6 \end{bmatrix} \qquad \text{and} \qquad \begin{bmatrix} 2 \\ 4 \end{bmatrix} \times (-1) = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$$



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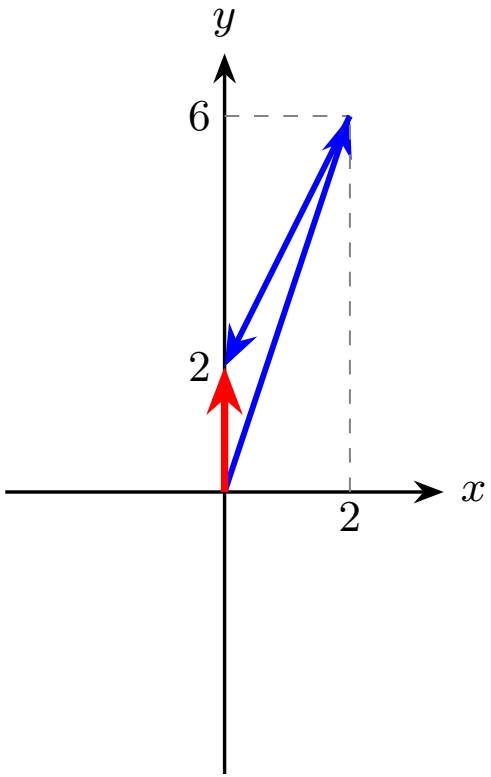
❸ **Vector Addition**: Sum the scaled vectors to find the transformed vector:

$$\begin{bmatrix} 2 \\ 6 \end{bmatrix} + \begin{bmatrix} -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

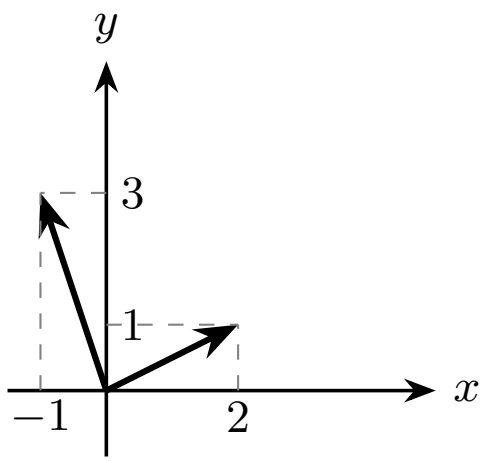


# Linear Transformation

**Understand $Ax$ geometrically**

- Consider $\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$ and $\boldsymbol{x} = \begin{bmatrix} 2 \\ \frac{1}{2} \end{bmatrix}$, we have

$$\boldsymbol{Ax} = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ \frac{1}{2} \end{bmatrix} = 2 \times \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{1}{2} \times \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} -\frac{1}{2} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} \frac{7}{2} \\ \frac{7}{2} \end{bmatrix}$$



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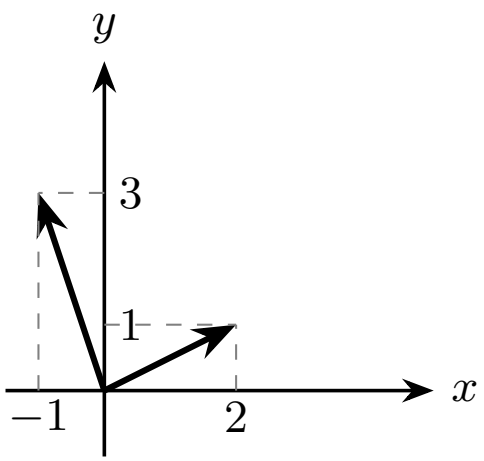


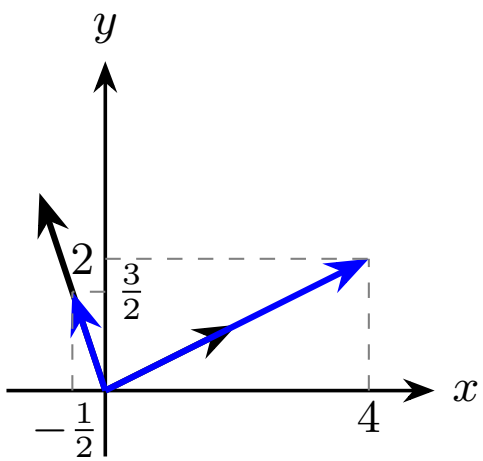

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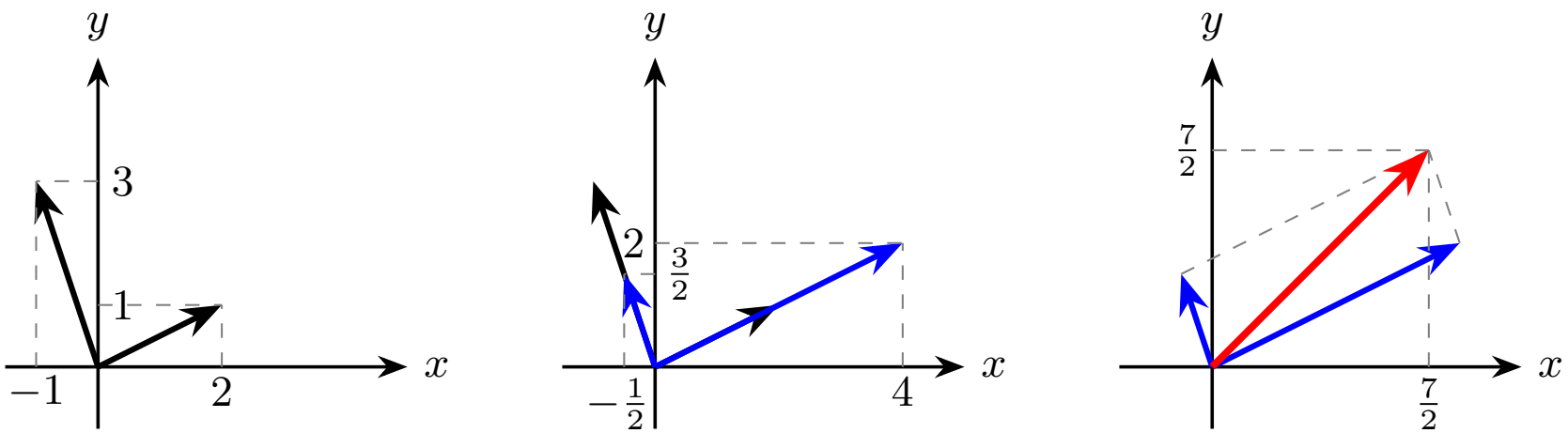

# Matrix Multiplication

- If $\boldsymbol{C} = \boldsymbol{AB}$, then the element $c_{ij}$ is defined by the inner product:

$$c_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}$$

- **Example**: Define two matrices:

$$\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

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  To find each element $c_{ij}$, we take the **inner product** of the $i$-th row of $\boldsymbol{A}$ and the $j$-th column of $\boldsymbol{B}$:

  - For $c_{11}$: $(1 \times 5) + (2 \times 7) = 5 + 14 = \mathbf{19}$
  - For $c_{12}$: $(1 \times 6) + (2 \times 8) = 6 + 16 = \mathbf{22}$
  - For $c_{21}$: $(3 \times 5) + (4 \times 7) = 15 + 28 = \mathbf{43}$
  - For $c_{22}$: $(3 \times 6) + (4 \times 8) = 18 + 32 = \mathbf{50}$

- Python programming: `C = A @ B`

# Matrix Multiplication

- **Complexity**: $\mathcal{O}(n^3)$ operations for two $n \times n$ matrices
- Why is it $\mathcal{O}(n^3)$?
  - There are $n^2$ total elements in the resulting matrix $(2 \times 2 = 4)$
  - To calculate each element, we perform $n$ multiplications ($2$ in this case)
  - Total operations $\approx n^2 \times n = n^3$

## Matrix Multiplication

Given $\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\boldsymbol{X} = \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix}$, compute $\boldsymbol{AX}$.

# Matrix Multiplication

Given $\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\boldsymbol{X} = \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix}$, compute $\boldsymbol{AX}$.

$$\begin{aligned} \boldsymbol{AX} &= \begin{bmatrix} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{bmatrix} \\ &= \begin{bmatrix} 21 & 24 & 27 \\ 47 & 54 & 61 \end{bmatrix} \end{aligned}$$

# Determinant

- **Definition**: The determinant is a scalar value that can be computed from the elements of a square matrix, i.e., $\det : \mathbb{R}^{n \times n} \to \mathbb{R}$
- **Geometric Significance**: It represents the volume scaling factor of the linear transformation described by the matrix
- **Singularity**: If the determinant is zero, the matrix is singular, meaning the system of equations $\boldsymbol{Ax} = \boldsymbol{b}$ may have no unique solution
- **Python**: `np.linalg.det(A)`

# Determinant

Matrix of size $2 \times 2$:

- The determinant of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$$

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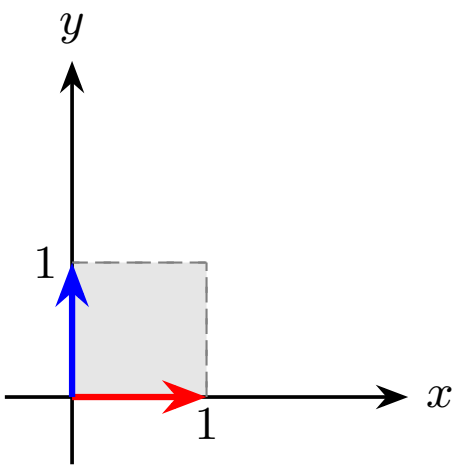
- Intuitive understanding of $\det\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 1$
  - Draw column vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, the area is $1$



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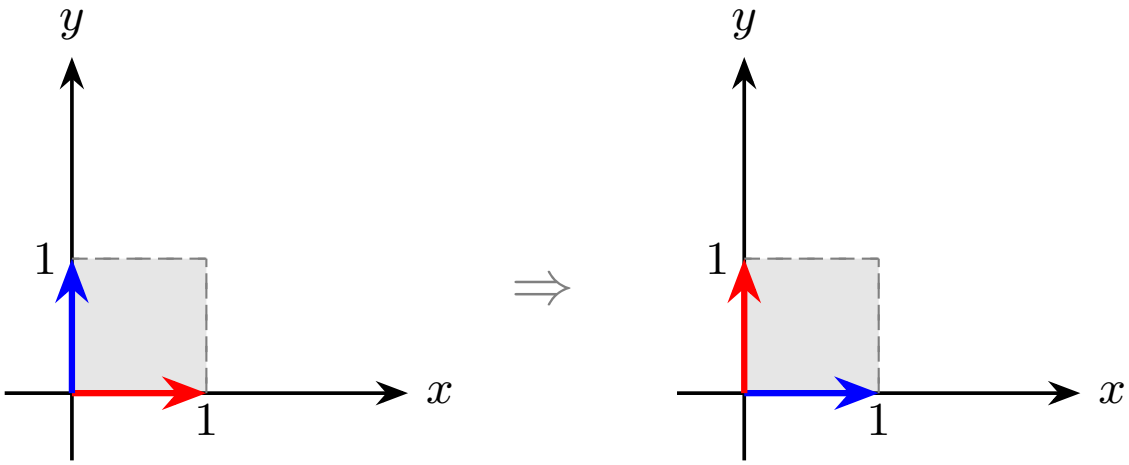
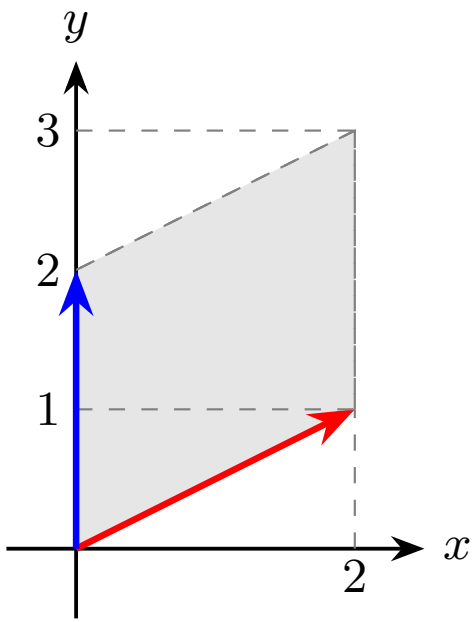
- Intuitive understanding of $\det\left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\right) = 0 - 1 = -1$
  - Draw column vectors $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, the area is $1$ (reversed direction)



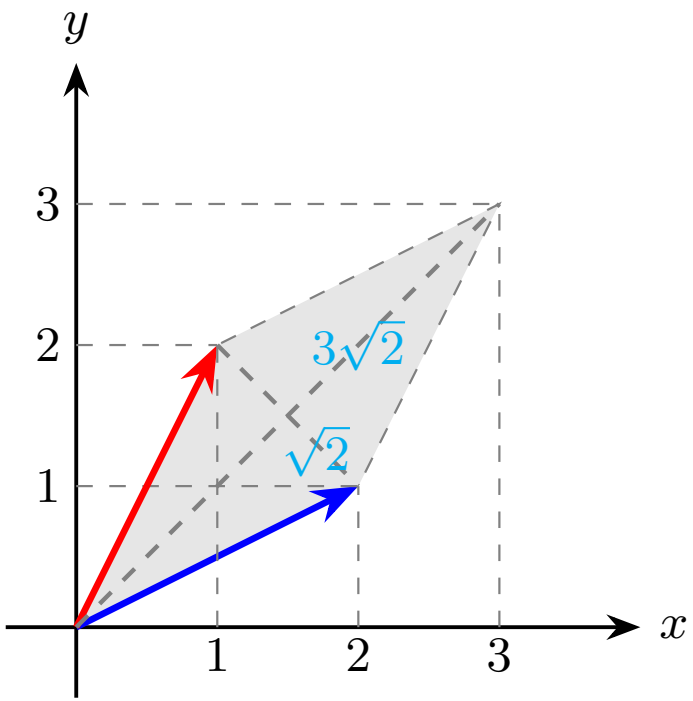
# Determinant

- Intuitive understanding of the determinant of $\boldsymbol{A} = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$
  - Draw column vectors $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$
  - Area is $4$, so $\det(\boldsymbol{A}) = 4$



# Determinant

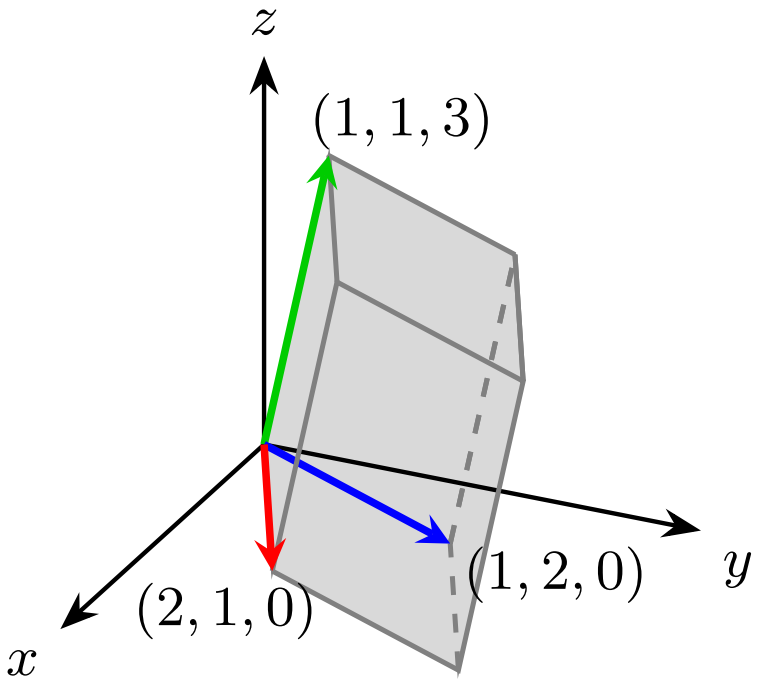
- Intuitive understanding of the determinant of $\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$
  - Draw column vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$
  - Area is $\frac{1}{2} \times 3\sqrt{2} \times \sqrt{2} = 3$ (reversed direction), so $\det(\boldsymbol{A}) = -3$



# Determinant

Matrix of size $3 \times 3$:

- Intuitive understanding of the determinant of $\boldsymbol{A} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$
  - Draw column vectors $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$, volume $\det(\boldsymbol{A}) = 9$



## Quick Summary

**Monday's Class**:

- A system of linear equations
- Linear transformation
- Matrix multiplication
- Determinant of square matrices